

AI-Powered Predictive Maintenance using Digital Twin

1. Project Overview

This system utilizes a Digital Twin framework powered by unsupervised machine learning (Autoencoders) to monitor industrial assets. By learning the "healthy" baseline of a machine, the system can detect subtle anomalies through reconstruction errors. The core innovation is the "What-If" Analysis engine, which allows operators to simulate environmental changes and see predicted outcomes before they happen.

2. System Architecture

Data Input: Real-time monitoring of Temperature, Vibration, and Load.

Digital Twin Core: An autoencoder model that mirrors the physical asset's behavior.

Health Indexing: Conversion of model deviations into a 0–100% health score.

Predictive Engine: Calculates Remaining Useful Life (RUL) based on degradation trends.

Interactive Simulation: Sliders to adjust parameters for scenario testing

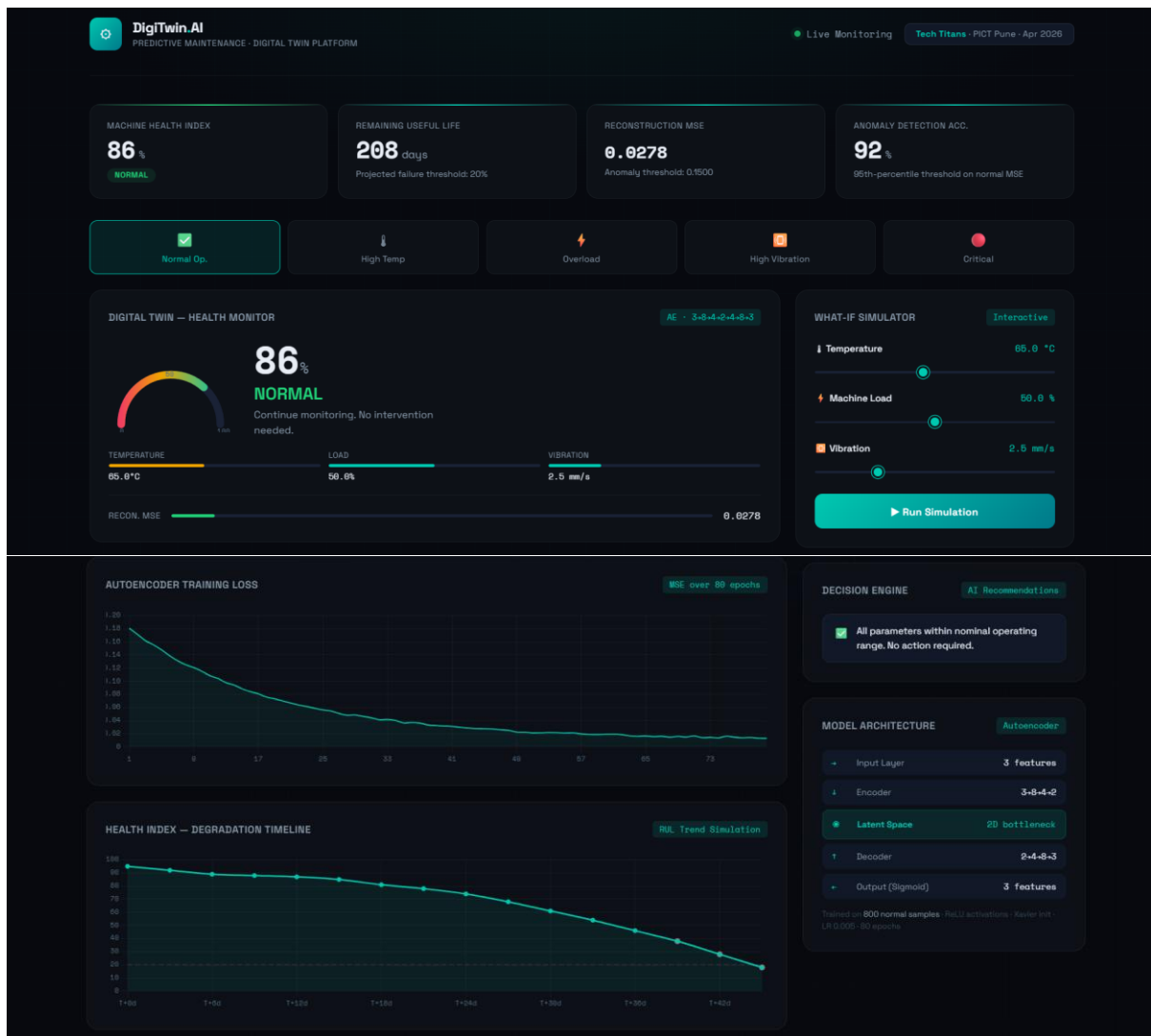
3. Key Features

Proactive Maintenance: Shifts from "Fix when broken" to "Fix before failure."

Explainable AI: Uses clear visual logs and warnings (Warn/Critical) to justify recommendations.

Actionable Insights: Suggests cooling or load reduction based on real-time simulation results.

3. Visual Representation





DigiTwin.AI - AI-Powered Predictive Maintenance using Digital Twin. **Tech Titans** - PICT Pune - April 2026
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